Yatharth Goyal

Software Engineer | Backend Development & Distributed Systems

Jaipur, India | yatharthgoyal657@gmail.com

GitHub | LinkedIn | Portfolio

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL

Backend & Databases: Node.js, Express, PostgreSQL, Redis, BullMQ, FastAPI, REST APIs, PostGIS

Tools & Delivery: React, Vite, Git, Docker, GitHub Actions, pytest, Playwright, AWS SAM

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Relational Systems, Asynchronous Workflows

PROFESSIONAL EXPERIENCE

Grras Solutions Pvt. Ltd. | *Jaipur, India*

Data Science & Machine Learning Intern | June 2026 – July 2026

- Engineered and delivered a comprehensive Data Science and Machine Learning project, successfully meeting all technical requirements and strict deployment deadlines.
- Applied advanced machine learning concepts through hands-on development, adapting quickly to new technologies and exceeding engineering performance expectations.

ENGINEERING PROJECTS

AegisED | *PostgreSQL, FastAPI, Python, REST APIs*

Case study: AegisED architecture | GitHub: github.com/YT-PRO12/AegisED

- Engineered a transactional patient intake and allocation system utilizing PostgreSQL row locks to guarantee data integrity during concurrent updates.
- Developed secure workflow transitions enforced by strict role-based access control (RBAC) and comprehensive system audit trails.
- Integrated a FastAPI-based machine learning microservice for medical case analysis, achieving a 0.7893 macro-F1 score across 900 test cases.

EvalForge | *Node.js, Redis, BullMQ, PostgreSQL*

Case study: EvalForge system design (private source)

- Architected an asynchronous grading engine utilizing BullMQ and Redis to ensure idempotent job processing and durable PostgreSQL persistence.
- Implemented complex backend validation logic with 15 comprehensive local checks, successfully linking automated rubric scores to exact textual evidence.

Grid Flex | *Python, SciPy, HiGHS, AWS SAM*

Live Demo: chimerical-praline-ff8e07.netlify.app | GitHub: github.com/YT-PRO12/Grid-Flex

- Developed an energy scheduling optimization engine using Python, SciPy, and HiGHS solvers to balance operational costs and carbon emissions.
- Authored 52 comprehensive regression test suites to validate complex power balance and battery state scenarios.
- Designed and configured AWS Serverless Application Model (SAM) templates for scalable cloud infrastructure deployment.

EduPredict AI | *React, Vite, FastAPI, SQLite*

Live Demo: edu-predict-ai-64gq.vercel.app | GitHub: github.com/YT-PRO12/EduPredictAI

- Developed a React/Vite student-success dashboard connected to FastAPI for assessments, risk insights, academic history, and intervention workflows.
- Implemented server-generated assessment IDs and student-scoped intervention persistence to protect submissions and support-plan updates.

EDUCATION & CERTIFICATIONS

JECRC Foundation

B.Tech in Information Technology | CGPA: 9.00/10.00 | *Expected Graduation: 2028*

Certifications:

- **Microsoft & LinkedIn Learning:** Azure AI Engineer Associate (AI-102) Cert Prep; Azure AI Fundamentals (AI-900) Cert Prep; Azure AI Essentials; Career Essentials in Generative AI
- **Google Cloud & Simplilearn:** Introduction to Generative AI
- **Scaler:** Java Course - Mastering the Fundamentals
- **IBM & Analytics Vidhya:** Python 101 (IBM); Introduction to Python (Analytics Vidhya)